Classical entanglement of purification is nonadditive with three outcomes per party

Ying Xie
Kennesaw State University
yxie2@kennesaw.edu

Abstract

We exhibit an explicit rational 3×3 classical probability table whose regularized von Neumann entanglement of purification is strictly smaller than its one-copy value. In nats, we prove $E_{\mathrm{P}} \geq 0.07163132826$ and $E_{\mathrm{P}}^{\infty} \leq 0.071592778856911368$. For every $n \geq 10^{12}$ we also prove $E_{\mathrm{P}}(\rho^{\otimes n})/n < 0.071611$ by a direct classical construction. It specifies an exact purification through a terminating finite search; it is not a practical small-copy witness. A support-degree rank reduction handles arbitrary finite ancillas and reduces the one-copy problem to two phases. Rational entropy quadrature and elementary minima of three-cosine remainders certify the lower bound. A two-component covering argument supplies the upper bound without invoking the Chen–Winter theorem. The gap persists for strictly positive tables. We also prove an obstruction to the same ensemble strategy for binary classical–quantum targets; binary additivity remains open. Standard-library verifiers replay the finite arithmetic. The accompanying research record documents the Recursive Reasoning Exploration run and subsequent refinements. An accompanying three-view visualization guides the reader through the discovery route, proof dependencies, and provenance of the mathematical inputs.

1 A ternary classical separation

For a probability table $P = (P_{ij})$, write

$$\rho_P = \sum_{ij} P_{ij} |i,j\rangle\langle i,j|, \qquad E_{\mathrm{P}}(P) = E_{\mathrm{P}}(\rho_P).$$

The entanglement of purification is $E_{\mathrm{P}}(\rho) = \inf_{\psi} S(AA')_{\psi}$, with the infimum over purifications $\psi_{AA'BB'}$ with arbitrary finite ancillas. All logarithms here are natural, and all entropy values are in nats. Tensoring purifications gives subadditivity, so

$$E_{\mathrm{P}}^{\infty}(\rho) = \lim_{n\to\infty} \frac{E_{\mathrm{P}}(\rho^{\otimes n})}{n} = \inf_{n\geq 1} \frac{E_{\mathrm{P}}(\rho^{\otimes n})}{n}.$$

Theorem 1. *Let*

$$P = \frac{1}{2482} \begin{pmatrix} 64 & 36 & 0 \\ 0 & 4 & 16 \\ 2025 & 256 & 81 \end{pmatrix}. \tag{1}$$

Then

$$E_{\mathrm{P}}(P) \geq 0.07163132826, \qquad E_{\mathrm{P}}^{\infty}(P) \leq 0.071592778856911368. \tag{2}$$

Consequently

$$E_{\mathrm{P}}(P) - E_{\mathrm{P}}^{\infty}(P) \geq 0.000038549403088632 > 0.000038. \tag{3}$$

Moreover, for every integer $n \geq 10^{12}$,

$$E_{\mathrm{P}}(\rho_P^{\otimes n})/n < 0.071611 < E_{\mathrm{P}}(P). \tag{4}$$

Corollary 2. *Let* J *be the* 3×3 *table with every entry* $1/9$. *For every* $0 < \varepsilon \leq 10^{-12}$, *the strictly positive table* $P_\varepsilon = (1-\varepsilon)P + \varepsilon J$ *satisfies*

$$E_{\mathrm{P}}(P_\varepsilon) - E_{\mathrm{P}}^{\infty}(P_\varepsilon) > 0.000022.$$

The proof of the theorem is in Sections 2–5; Section 6 proves the corollary. The theorem supplies an explicit sufficient tensor-power length; the least violating length and a two-copy violation remain unknown. Neither entropy is claimed to have been evaluated exactly. If $d_{\min}$ is the least square classical alphabet size admitting such a regularization gap, allowing zero probabilities, the theorem gives $d_{\min} \in \{2, 3\}$: the one-outcome case is product. It does not prove that $d_{\min} = 3$ or settle the binary conjecture of [3]. Minimal rectangular alphabet pairs are also not classified.

Terhal et al. introduced this quantity and its operational interpretation [1]. Chen and Winter state that $E_{\mathrm{P}}^{\infty} - S$ is convex in [2, Corollary 4]. Recent work proves classical two-qubit Rényi nonadditivity below order one [3], and von Neumann nonadditivity for separable Werner states [4]. Those Werner states are not diagonal in a local product basis; the corresponding classical von Neumann question is explicitly left open in [4]. Theorem 1 supplies a classical example with three outcomes per party.

Scope and priority. The result concerns classical states at von Neumann order, rather than all separable states or every Rényi order. A targeted literature search on 3 November 2026 checked the primary texts of [3, 4], their related works, and newer Rényi-2 results [6, 7]. No prior classical von Neumann counterexample was identified in those sources. This bounded search does not establish worldwide priority or exclude unpublished and concurrent work; its queries and source snapshots are retained in the supplement. The contribution asserted here is the explicit certified ternary example and its quantified construction.

2 Removing unrestricted ancillary dimensions

For compatible complex block matrices $M = (M_{ij})$ of arbitrary finite block sizes, one has

$$E_{\mathrm{P}}(P) = \inf_{\|M_{ij}\|_F^2 = P_{ij}} S(MM^*). \tag{5}$$

Indeed, the coefficient matrix of any purification supplies feasible blocks, with the same Schmidt spectrum across $AA' : BB'$. Conversely, any feasible M defines the purification

$$\sum_{ijab} (M_{ij})_{ab} |i\rangle_A |i, a\rangle_{A'} |j\rangle_B |j, b\rangle_{B'}.$$

Tracing the ancillary registers kills all unequal physical labels and leaves ρ_P. The coefficient matrix across $AA' : BB'$ is an isometric embedding of M, preserving its singular values. Padding handles different block sizes. These two constructions prove (5).

Lemma 3 (Fixed-trace rank reduction)**.** *Let a positive semidefinite matrix satisfy* s *real linear constraints whose span fixes its positive trace. For any concave objective on the feasible set, it has a feasible replacement of rank at most* $\lfloor \sqrt{s} \rfloor$ *and no greater objective value.*

Proof. Write the matrix as $K = VV^*$ of rank r, with V of full column rank. If $r^2 > s$, the real vector space of Hermitian $r \times r$ matrices contains a nonzero H with every constrained functional vanishing on VHV^*. Since $\mathrm{Tr}(V^*VH) = 0$ and $V^*V > 0$, H is indefinite. The segment $V(I+tH)V^*$ has finite positive semidefinite endpoints on both sides of zero, each of smaller rank. The original matrix is a convex combination of the endpoints; a concave objective is no larger at one of the two endpoints. Repeating this step terminates at $r^2 \le s$. □

Lemma 4 (Ancillary dimensions from support degrees)**.** *Suppose row i of P has support size d_i and column j has support size e_j. After zero rows and columns are removed, the infimum in (5) is attained with internal row dimensions at most $\lfloor\sqrt{d_i}\rfloor$ and internal column dimensions at most $\lfloor\sqrt{e_j}\rfloor$. If every support degree is at most three, a scalar-phase amplitude attains $E_{\mathrm{P}}(P)$.*

Proof. First reduce the column spaces. Stack column block j of M as Y_j and put $K_j = Y_jY_j^*$. Then $MM^* = \sum_j K_j$. On the direct sum of incident row spaces, K_j obeys the e_j constraints $\mathrm{Tr}(\Pi_i K_j) = P_{ij}$, whose sum fixes its trace. A zero block has zero support and may be omitted. With the other columns fixed, $S(\sum_j K_j)$ is concave. Lemma 3 reduces K_j to rank at most $\lfloor\sqrt{e_j}\rfloor$. Refactoring preserves every block norm and bounds that column's internal dimension. Do this for every column.

Now fix these reduced column spaces and use M^*M, which has the same nonzero spectrum. Each stacked row X_i contributes $G_i = X_i^*X_i$, constrained by the d_i traces $\mathrm{Tr}(\Omega_j G_i) = P_{ij}$. Apply Lemma 3 again and factor the reduced G_i. This factor has fewer rows but acts on the same direct sum of column spaces. Thus the row reduction preserves the column bounds from the first pass as well as every block norm.

Every feasible matrix therefore has a replacement of uniformly bounded size and no greater entropy. After padding, the feasible set is a finite product of Frobenius spheres. Compactness and continuity give attainment. Degrees at most three force each internal dimension to be one. □

These are internal block dimensions; the outcome-copy flags in the purification remain. The scalar conclusion holds for every 3×3 table, including full support. It is applied to one copy only: support degrees grow under tensor products, so it does not restrict collective purifications to scalar amplitudes.

3 The complete two-phase problem

Set $W = 2482$ and

$$B(z,w) = \begin{pmatrix} 8 & 6 & 0 \\ 0 & 2 & 4 \\ 45 & 16z & 9w \end{pmatrix}, \qquad G(z,w) = B(z,w)B(z,w)^*/W.$$

Corollary 5 (Two-phase reduction)**.** *For the table (1), the unrestricted one-copy problem is*

$$E_{\mathrm{P}}(P) = \min_{z,w\in\mathbb{T}} S(G(z,w)). \tag{6}$$

Proof. Lemma 4 reduces every feasible purification to scalar amplitudes without increasing entropy. The support has seven edges; the five edges $(1,1),(1,2),(2,2),(2,3),(3,1)$ form a spanning tree on its six vertices. Successive row and column phase changes make these entries positive without changing singular values. The remaining $7-6+1 = 2$ phases are z, w, giving exactly the displayed family. □

Write $E_k = e_k(G)$ for its elementary symmetric coefficients. Its trace is one, so

$$\mathcal{D}_E(t) = \det(tI + G) = t^3 + t^2 + E_2 t + E_3. \tag{7}$$

Direct expansion gives

$$W^2 E_2 = 144160 - 34560(z + z^{-1}) - 1152(z/w + w/z), \tag{8}$$

$$W^3 E_3 = |1080 + 144w - 512z|^2. \tag{9}$$

For example, the diagonal entries of BB^* are $(100, 20, 2362)$, and its upper off-diagonal entries are 12, $360 + 96\overline{z}$, and $32\overline{z} + 36\overline{w}$. These give (8); expanding $\det B$ gives (9). Since $\Re z, \Re(z/w) \le 1$ and $1080 > 144 + 512$, one obtains the uniform rational bounds

$$E_2 \ge a_2 := \frac{72736}{W^2}, \qquad E_3 \ge a_3 := \frac{424^2}{W^3}. \tag{10}$$

In particular, every feasible G is positive definite.

4 A global entropy certificate

The lower-bound proof has two parts: an integral converts entropy into monotone functions of (E_2, E_3), and eight rational affine certificates control the remaining coefficient rectangle. Their sum-of-squares identities establish a baseline bound; exact minima of the positive remainders strengthen it. The certificate data are finite rational numbers; their arithmetic bounds are checked with outward rounding. The integral, quadrature, and rectangle arguments below establish why those finite checks imply a uniform entropy bound for every feasible Gram matrix.

4.1 Entropy integral and tails

Choose the reference spectrum $r = (247/250, 1/100, 1/500)$, with sum one, and put $\mathcal{D}_r(t) = \prod_j (t + r_j)$ and $S_r = -\sum_j r_j \log r_j$. Integrating $\log(t + x)$ by $(t + x)\log(t + x) - (t + x)$ gives

$$S(G) = S_r + \int_0^\infty \log \frac{\mathcal{D}_E(t)}{\mathcal{D}_r(t)}\, dt. \tag{11}$$

Equal dimension and equal trace cancel the boundary terms at infinity; the boundary at zero is the entropy difference. In particular, $\log(\mathcal{D}_E(t)/\mathcal{D}_r(t)) = O(t^{-2})$ as $t \to \infty$. The positive determinant bound in (10) also bounds the integrand near zero. Thus the improper integral converges. Take $\tau = 10^{-8}$ and $T = 1000$. By (10), the small tail has the lower bound

$$\int_0^\tau \log(\mathcal{D}_E/\mathcal{D}_r)\, dt \ge \tau \log \frac{a_3}{\prod_j (\tau + r_j)}. \tag{12}$$

Let $R_2 = e_2(r)$, $R_3 = e_3(r)$, and $\delta_k = \max(R_k - a_k, 0)$ for $k = 2, 3$. Since $\mathcal{D}_r(t) \ge t^3$,

$$\mathcal{D}_E(t)/\mathcal{D}_r(t) \ge 1 - \delta(t), \qquad \delta(t) = \delta_2/t^2 + \delta_3/t^3.$$

The exact check gives $\delta(T) < 1$. The inequality $\log(1 - u) \ge -u/(1 - u)$ consequently gives

$$\int_T^\infty \log(\mathcal{D}_E/\mathcal{D}_r)\, dt \ge -\frac{\delta_2/T + \delta_3/(2T^2)}{1 - \delta(T)}. \tag{13}$$

4.2 Quadrature and the outer rectangle

For an actual positive semidefinite Gram,

$$\frac{d^4}{dt^4}\log\det(tI+G) = -6\,\mathrm{Tr}((tI+G)^{-4}) < 0.$$

Thus Simpson quadrature is a lower integral bound. Its sign can also be read from the Peano kernel for integral minus Simpson on $[0,1]$: $-s^3(2-3s)/72$ for $s \le 1/2$, and $-(1-s)^3(3s-1)/72$ for $s \ge 1/2$. Both are nonpositive. Rescaling and summing gives the required direction on every panel. The certificate uses 2546 rational panels and 5093 combined nodes. The reference integral is evaluated by its elementary antiderivative and bounded above. Together with (12) and (13), this gives a uniform bound

$$S(G) \ge c + \sum_t \omega_t \log \mathcal{D}_E(t), \qquad \omega_t > 0, \tag{14}$$

with an exact rational lower constant c.

Write $a = (a_2, a_3)$ and choose the exact upper corner

$$b = (0.01255565838,\ 0.000027858287).$$

Substituting (b_2, a_3) and (a_2, b_3) into the right side of (14) yields certified lower bounds greater than 0.07163919 and 0.07163993, respectively. That expression increases in both coefficients. Every feasible pair outside $[a,b]$ therefore has entropy greater than L_* in (20). Simpson's derivative sign is used only for actual Grams; artificial corner polynomials are substituted into the already established finite expression.

4.3 Eight rational sum-of-squares identities

Divide $[a_2, b_2]$ into two equal intervals and $[a_3, b_3]$ into four. For each resulting closed rectangle $[u, v]$ and each node t, let $p_- = \mathcal{D}_u(t)$ and $p_+ = \mathcal{D}_v(t)$. The stored rational logarithm bounds $l_- \le \log p_-$ and $l_+ \le \log p_+$ satisfy $l_- < l_+$ and $0 < p_- < p_+$. Concavity gives, throughout the rectangle,

$$\log \mathcal{D}_E(t) \ge l_- + \frac{l_+ - l_-}{p_+ - p_-}\big[t(E_2 - u_2) + (E_3 - u_3)\big]. \tag{15}$$

Indeed the linear interpolation of the lower endpoint bounds lies below the chord of the true logarithm, which lies below the logarithm itself. Substitution into (14) yields

$$S(G) \ge \beta + \kappa \cdot (E - u). \tag{16}$$

The entries of the weight vector κ are rounded downward; this preserves the inequality because $E - u$ is coordinatewise nonnegative.

For each rectangle the certificate supplies nonnegative rational multipliers m^-, m^+ and a rational symmetric 4×4 matrix Γ satisfying the Laurent-polynomial identity

$$\kappa \cdot E - \ell - m^- \cdot (E - u) - m^+ \cdot (v - E) = \eta(z,w)^* \Gamma \eta(z,w), \tag{17}$$
$$\eta(z,w) = (1, w, z, zw)^T, \qquad \ell = 0.07163 - \beta + \kappa \cdot u.$$

Every coefficient of (17) is checked exactly. Independent rational LDL elimination gives four positive pivots for each Γ. Thus the right side is nonnegative, including for complex monomials. On the rectangle the two multiplier terms are also nonnegative after rearrangement, giving $\kappa \cdot E \ge \ell$. Equation (16) then gives $S(G) \ge 0.07163$. The eight rectangles cover $[a,b]$, including its boundaries. Together with the outside bound and Lemma 4, this proves the baseline bound $E_{\mathrm{P}}(P) \ge L_0 = 0.07163$.

4.4 Retaining the positive remainder

The preceding argument discarded a strictly positive remainder. On its rectangle, (16) and (17) give

$$S(G) \geq L_0 + f(z,w), \qquad L_0 = 0.07163, \quad f(z,w) = \eta(z,w)^* \Gamma \eta(z,w).$$

By (8)–(9), the left side of the Laurent identity (17), hence also f, has the form

$$f(e^{i\theta}, e^{i\phi}) = c_0 + c_1 \cos\theta + c_2 \cos\phi + c_3 \cos(\theta - \phi), \qquad c_j \in \mathbb{Q}. \tag{18}$$

Its minimum is elementary. For fixed $x = \cos\theta$, minimizing over ϕ yields

$$c_0 + c_1 x - \sqrt{c_2^2 + c_3^2 + 2c_2c_3x}, \qquad -1 \leq x \leq 1. \tag{19}$$

The two phase terms combine into a complex number of the displayed modulus, so this is an equality. The function is convex, with derivative $c_1 - c_2c_3/\sqrt{c_2^2 + c_3^2 + 2c_2c_3x}$ where the radical is positive. Its endpoint values are $c_0 - c_1 - |c_2 - c_3|$ and $c_0 + c_1 - |c_2 + c_3|$. If $c_1c_2c_3 > 0$ and

$$x_* = \frac{(c_2c_3/c_1)^2 - c_2^2 - c_3^2}{2c_2c_3} \in [-1, 1],$$

one also includes the stationary value

$$c_0 - \frac{c_1(c_2^2 + c_3^2)}{2c_2c_3} - \frac{c_2c_3}{2c_1}.$$

The least candidate is the global minimum. If $c_1c_2c_3 = 0$, endpoints suffice. Zeros of the radical occur only at a boundary unless $c_2 = c_3 = 0$, already covered. All candidate values are rational.

The verifier derives the c_j from each rectangle's affine coefficients and multipliers. The least of the eight global remainder minima is

$$\frac{25386301346993002467567602489873221371}{191124052100000000000000000000000000000000000} > 0.00000132826.$$

This is the minimum for rectangle $(1, 0)$, using zero-based indices. The outside-rectangle bounds exceed 0.07163919, so the complete bound is

$$E_{\mathrm{P}}(P) \geq L_* := 0.07163132826. \tag{20}$$

The SOS factorizations remain independently checked representations of the same inequalities. The elementary minimization can also verify their affine remainders without constructing SOS matrices.

4.5 Exact logarithm enclosures

After extracting an integer power of two, write $1 \leq x \leq 2$ and $y = (x-1)/(x+1) \in [0, 1/3]$. Then

$$\log x = 2 \sum_{k=0}^{N-1} \frac{y^{2k+1}}{2k+1} + R_N, \qquad 0 \leq R_N \leq \frac{9}{4(2N+1)3^{2N+1}}.$$

The generator uses 32 terms with integer interval arithmetic. The separate verifier uses 40 terms and directly rounded powers at scale 10^{36}. After k multiplication steps the lower power differs from y^k by at most $k\,10^{-36}$; each rounded odd summand loses less than $3\,10^{-36}$. Adding these errors and the displayed remainder gives rational logarithm enclosures. Signed multiples of $\log 2$ use the appropriate endpoints. The verifier checks each stored lower endpoint against its separately computed lower enclosure, including all chord endpoints and tail constants.

5 A direct classical upper bound

Chen–Winter's inequality [2, Theorem 3] states that, for an ensemble $\rho = \sum_h \lambda_h \rho_h$ of possibly mixed bipartite states,

$$E_{\mathrm{P}}^{\infty}(\rho) \le \sum_h \lambda_h E_{\mathrm{P}}^{\infty}(\rho_h) + S(\rho) - \sum_h \lambda_h S(\rho_h). \tag{21}$$

Here S is joint-state entropy. Their proof outlines an argument using Ahlswede–Winter operator sampling and Winter's 2004 Theorem 2, cited there. The equivalent convexity statement is [2, Corollary 4]. We give a direct classical construction for the ensemble used here, including explicit finite-copy constants. Thus (21) supplies context but is not a premise of Theorem 1 or Corollary 2.

5.1 The rational ensemble and its rate

List the seven support edges of P in row-major order and assign

$$(\alpha_e)_e = \left(\frac{139}{1000}, \frac{27}{200}, \frac{117}{500}, \frac{157}{500}, \frac{71}{500}, \frac{18}{125}, \frac{19}{125}\right). \tag{22}$$

Set $c_e = (1-\alpha_e^2)/(1+\alpha_e^2)$ and $s_e = 2\alpha_e/(1+\alpha_e^2)$, so $c_e^2 + s_e^2 = 1$. Starting from $B(1,1)$, multiply each nonzero entry by c_e and s_e to form rational matrices A_0, A_1, respectively. Negate entry $(3,3)$ of A_0 only, and set

$$(Q_h)_{ij} = (A_h)_{ij}^2/W, \quad K_h = A_h A_h^T/W, \quad \lambda_h = \operatorname{Tr} K_h, \qquad h = 0, 1. \tag{23}$$

Exactly, $Q_0 + Q_1 = P$ and $\lambda_0 + \lambda_1 = 1$, with positive weights. Put $q_h = Q_h/\lambda_h$, $D_h = A_h/\sqrt{W\lambda_h}$, and $\sigma_h = D_h D_h^*$. The amplitudes D_h, equipped with local outcome-copy flags, purify the classical tables q_h.

Define the information and ensemble rate

$$I_c = \sum_{h,e} \lambda_h q_h(e) \log \frac{q_h(e)}{P(e)} = H(P) - \sum_h \lambda_h H(q_h), \tag{24}$$

$$R_c = I_c + \sum_h \lambda_h S(\sigma_h) = H(P) + \sum_h [\mathcal{S}(K_h) - \mathcal{H}(Q_h)], \tag{25}$$

where $\mathcal{S}(X) = -\operatorname{Tr} X \log X$ and $\mathcal{H}(Q) = -\sum_e Q(e) \log Q(e)$ include unnormalized arguments. The $-\lambda_h \log \lambda_h$ terms cancel in the last equality.

For each rational K_h the certificate supplies three rational approximate eigenpairs $(\mu, \mathbf{x})$ with radius $\xi = 10^{-30}$ and

$$\|K_h \mathbf{x} - \mu \mathbf{x}\|_2^2 \le \xi^2 \|\mathbf{x}\|_2^2, \qquad \mathbf{x} \ne 0.$$

Spectral decomposition places an eigenvalue in every interval $[\mu - \xi, \mu + \xi]$. The intervals are disjoint subsets of $(0,1)$, so they contain all three eigenvalues. For $\nu \in [l, u] \subset (0,1)$, $-\nu \log \nu \le -u \log l$ bounds both nonnegative factors separately. Exact rational probabilities and directed logarithm bounds therefore give

$$R_c \le U_* := \frac{8949097357113921}{125000000000000000} = 0.071592778856911368. \tag{26}$$

This is an arithmetic statement about an explicit ensemble. The next construction supplies its interpretation as an asymptotic upper bound.

5.2 Entropy estimates

We use entropy concavity and the upper mixing bound

$$S\left(\sum_j p_j \tau_j\right) \le H(p) + \sum_j p_j S(\tau_j). \tag{27}$$

Indeed, $\bar{\tau} = \sum_j p_j \tau_j \ge p_j \tau_j$, so monotonicity of the matrix logarithm gives $D(\tau_j \| \bar{\tau}) \le -\log p_j$. Summing proves (27). Logarithm monotonicity follows from its resolvent integral and inverse-order monotonicity; singular cases follow by regularization on the joint support.

For density matrices of dimension d at trace distance $\Delta = \frac{1}{2}\|\rho - \sigma\|_1$, we also use

$$|S(\rho) - S(\sigma)| \le \Phi(\Delta, d), \quad \Phi(\Delta, d) = \Delta \log d + (1+\Delta)\log(1+\Delta) - \Delta \log \Delta. \tag{28}$$

To prove this, write $\rho - \sigma = X_+ - X_-$ with $\operatorname{Tr} X_+ = \operatorname{Tr} X_- = \Delta$. The state $(\rho + X_-)/(1+\Delta) = (\sigma + X_+)/(1+\Delta)$ has two mixture representations. Apply concavity to one and (27) to the other, then interchange them. The entropy difference of the normalized $X_\pm$ is at most $\log d$, giving (28). The right side increases with Δ; zero distance follows by continuity.

5.3 Classical covering and exact probability correction

Lemma 6. *Let $p = \lambda_0 q_0 + \lambda_1 q_1$ be a rational mixture of two rational probability tables positive on the same finite support $\mathcal{E}$. Let D_h have d_A rows and satisfy $|(D_h)_{xy}|^2 = q_h(x,y)$; set $\sigma_h = D_h D_h^*$ and define I_c, R_c as in (24)–(25), with P replaced by p. Choose rational $1 < r_c < s_c$, $0 < \epsilon_c < 1$, $\zeta > 0$, and positive δ_c, γ_c, b_c, such that*

$$\log r_c - I_c \ge \delta_c, \qquad \log(s_c/r_c) \ge \gamma_c,$$

and the range of $\log(q_h(e)/p(e))$ has width at most b_c. Put $\Delta_c = \sqrt{\epsilon_c}$ and

$$B_n = 2e^{-2n\delta_c^2/b_c^2} + e^{-n\gamma_c/2}.$$

If $B_n/\epsilon_c + 1/(4n\zeta^2) < 1$, an exact purification satisfies

$$\begin{aligned}\frac{E_{\mathrm{P}}(p^{\otimes n})}{n} &\le R_c + \log s_c - I_c + (\zeta + \Delta_c)\log d_A \\ &\quad + \frac{\log 2 + (1+\Delta_c)\log(1+\Delta_c) - \Delta_c \log \Delta_c}{n}.\end{aligned} \tag{29}$$

Consequently $E_{\mathrm{P}}^{\infty}(p) \le R_c$.

Proof. Draw $M = \lceil s_c^n \rceil$ independent strings $H_j \in \{0,1\}^n$ with independent coordinates distributed as λ. For each string let $q_{h^n} = \bigotimes_i q_{h_i}$ and $D_{h^n} = \bigotimes_i D_{h_i}$. Write $\bar{q} = M^{-1}\sum_j q_{H_j}$, whose expectation is $p^{\otimes n}$.

For fixed e^n, the likelihood ratio $Z_{H_j} = q_{H_j}(e^n)/p^{\otimes n}(e^n)$ has mean one. Truncate it at r_c^n, obtaining Z'. Then $\operatorname{Var}(Z') \le \mathbb{E}(Z')^2 \le r_c^n$. Cauchy–Schwarz bounds the expected fluctuation of its sample mean by $\sqrt{r_c^n/M}$. Restoring the discarded sample mass and the missing expectation contributes twice the tail mass. Summing over e^n gives

$$\mathbb{E}\|\bar{q} - p^{\otimes n}\|_1 \le 2\tau_n + \sqrt{r_c^n/M}, \tag{30}$$

where τ_n is the probability that the sum of n independent information densities exceeds $n \log r_c$, under the joint law $\Pr(H = h, E = e) = \lambda_h q_h(e)$. For a variable X in an interval of width b_c, the second derivative of $\log \mathbb{E} e^{\upsilon X}$ is a variance under a tilted law, at most $b_c^2/4$. Integrating twice gives $\log \mathbb{E} e^{\upsilon(X - \mathbb{E}X)} \le \upsilon^2 b_c^2/8$. Markov's inequality, optimized at $\upsilon = 4\delta_c/b_c^2$, yields $\tau_n \le e^{-2n\delta_c^2/b_c^2}$. Thus (30) is at most B_n.

The nM labels are independent. If f_0 is their fraction of zeros, then $\mathbb{E} f_0 = \lambda_0$ and $\mathrm{Var}(f_0) \le 1/(4nM)$. Chebyshev gives

$$\Pr\{|f_0 - \lambda_0| > \zeta\} \le \frac{1}{4nM\zeta^2} \le \frac{1}{4n\zeta^2}.$$

Together with Markov applied to (30), the lemma's hypothesis gives a codebook satisfying both

$$\|\bar{q} - p^{\otimes n}\|_1 \le \epsilon_c, \qquad |f_0 - \lambda_0| \le \zeta. \tag{31}$$

The frequency condition matters because the two component entropies need not agree: their average per copy in this codebook is at most $\sum_h \lambda_h S(\sigma_h) + \zeta \log d_A$.

Concatenate the matrices horizontally, defining $\mathsf{A}_{x,(j,y)} = M^{-1/2} (D_{H_j})_{x,y}$. It has norm one and

$$\mathsf{A}\mathsf{A}^* = M^{-1} \sum_j D_{H_j} D_{H_j}^*, \qquad S(\mathsf{A}\mathsf{A}^*) \le \log M + n \left[\sum_h \lambda_h S(\sigma_h) + \zeta \log d_A \right]$$

by tensor entropy additivity and (27). Since $\bar{q} > 0$ on $\mathcal{E}^n$, define the exact correction

$$\mathsf{B}_{x,(j,y)} = \mathsf{A}_{x,(j,y)} \sqrt{\frac{p^{\otimes n}(x,y)}{\bar{q}(x,y)}}.$$

Each physical block now has squared norm $p^{\otimes n}(x,y)$. The state

$$\sum_{x,y,j} \mathsf{B}_{x,(j,y)} |x\rangle_A |x\rangle_{A'} |y\rangle_B |y, j\rangle_{B'}$$

is therefore an exact target purification, with Schmidt matrix B up to local isometries. The normalized amplitude vectors have overlap $\sum_{e^n} \sqrt{\bar{q}(e^n) p^{\otimes n}(e^n)} \ge 1 - \epsilon_c/2$, so their trace distance is at most Δ_c. Partial trace preserves this bound. Both left Grams have dimension d_A^n; the codebook label is on Bob's side. Apply (28) and $\log \lceil s_c^n \rceil \le n \log s_c + \log 2$ to obtain (29).

All tests in (31) are rational. Ordering all ordered M-tuples of binary strings lexicographically and choosing the first accepted tuple specifies a terminating finite search. No efficiency is asserted.

For the asymptotic conclusion, the parameters must also vary: keeping them fixed would retain the positive overhead in (29). Let ζ, Δ_c tend to zero, choose rational r_c, s_c with $I_c < \log r_c < \log s_c$ arbitrarily close to I_c, and take n sufficiently large. This proves the asymptotic assertion. □

5.4 Explicit copy count and quantitative scope

For the ensemble (23), use

$$n_0 = 10^{12}, \quad r_c = 5009/5000, \quad s_c = 1001801/1000000, \quad d_A = 3,$$
$$\delta_c = 3/200000, \quad \gamma_c = 9/10000000, \quad b_c = 2,$$
$$\epsilon_c = 10^{-16}, \quad \Delta_c = 10^{-8}, \quad \zeta = 1/500000.$$

Directed logarithm bounds check every hypothesis. Numerically, $I_c \approx 0.00178337200839$; its exact enclosures and the information-density width bound are in the certificate report. At $n = n_0$, both exponents in B_n are at least 100. Since $e > 8/3$, one has $B_n \le 3(3/8)^{100} < \epsilon_c/4$, while $1/(4n\zeta^2) = 1/16$. The joint acceptance probability is positive. Exact evaluation of (29), using (26) and a rational lower enclosure for I_c, gives

$$\frac{E_{\mathrm{P}}(P^{\otimes n})}{n} < 0.071610995205 < 0.071611 \qquad (n \ge 10^{12}). \tag{32}$$

For larger n the probability bounds improve and the $1/n$ terms decrease. Equations (20), (26) and Lemma 6 prove Theorem 1. Here $n_0 = 10^{12}$ counts copies of the state. The codebook instead contains $M = \lceil (1.001801)^n \rceil$ label strings, an exponential number. Positive acceptance probability proves that at least one such codebook passes both rational tests; finite lexicographic search specifies it without an efficiency guarantee. The copy count is sufficient, not minimal, and the codebook has not been enumerated. No practical small-copy witness is claimed.

Numerical context. The feasible phase pair $(z, w) = (1, -1)$ gives

$$S(B(1,-1)B(1,-1)^*/2482) \approx 0.0716379517027.$$

Its difference from U_* is about 0.00004517; global optimality is not asserted. Retaining the rational remainder improves the certified gap by about 3.57%, but the resulting gap is still only about 0.054% of the one-copy cost. A substantially stronger quantitative effect would require better bounds or a different table or ensemble.

6 Strictly positive tables

Take any scalar amplitude of P_ε. Choose an amplitude of P with the same phases on its seven support entries. The corresponding normalized pure states have overlap

$$\sum_{ij} \sqrt{P_{ij}(P_\varepsilon)_{ij}} \ge \sqrt{1-\varepsilon} \sum_{ij} P_{ij} = \sqrt{1-\varepsilon}.$$

Their trace distance is at most $\sqrt{\varepsilon}$, and partial trace cannot increase it. The left Grams have dimension three. Lemma 4 applies to every 3×3 table, including full support, so (28) gives

$$E_{\mathrm{P}}(P_\varepsilon) \ge L_* - \Phi(\sqrt{\varepsilon}, 3) \ge L_* - \Phi(10^{-6}, 3). \tag{33}$$

An elementary flagged-mixture argument suffices for the upper bound. For any finite ensemble $\sum_j p_j \rho_j$, attach orthogonal copies of the label j to both parties' ancillas in a superposition of component purifications. Tracing the flags gives the desired mixture, while the entanglement is $H(p)$ plus the average component entanglement. Taking infima gives $E_{\mathrm{P}}(\sum_j p_j \rho_j) \le H(p) + \sum_j p_j E_{\mathrm{P}}(\rho_j)$.

Expand $P_\varepsilon^{\otimes n}$ as a mixture over the n choices of P or J. The label entropy is $nh_2(\varepsilon)$, where $h_2(x) = -x \log x - (1-x)\log(1-x)$. If K is the number of P factors, then K is binomial with mean $(1-\varepsilon)n$. Local permutations group those factors, and J is product, so

$$E_{\mathrm{P}}(P_\varepsilon^{\otimes n}) \le nh_2(\varepsilon) + \mathbb{E} E_{\mathrm{P}}(P^{\otimes K}).$$

For any $a > 0$, the direct upper construction gives $E_{\mathrm{P}}(P^{\otimes k}) \le k(U_* + a)$ for every sufficiently large k. For smaller k, use $E_{\mathrm{P}}(P^{\otimes k}) \le k \log 3$. After division by n, the latter contribution is bounded by a fixed constant divided by n. Taking $n \to \infty$ and then $a \downarrow 0$ yields

$$E_{\mathrm{P}}^{\infty}(P_\varepsilon) \le (1-\varepsilon)U_* + h_2(\varepsilon) \le U_* + h_2(10^{-12}). \tag{34}$$

This argument also applies when ε is irrational and requires no ensemble theorem for regularized entanglement of purification. Directed logarithm bounds certify

$$L_* - \Phi(10^{-6}, 3) > 0.0716154141366, \qquad U_* + h_2(10^{-12}) < 0.0715927788856.$$

Their difference exceeds 0.000022, proving Corollary 2. Every entry is at least $\varepsilon/9 > 0$.

7 Mechanism, generality, and the binary obstruction

For a classical table Q put $F(Q) = E_{\mathrm{P}}(Q) - H(Q)$. Since the supplied purification of q_h has entropy $S(\sigma_h)$, (20) and (25) imply

$$F(P) - \sum_h \lambda_h F(q_h) \geq L_* - R_c \geq L_* - U_* > 0. \tag{35}$$

Thus the certificate proves a specific failure of convexity, without requiring exact component optima. The covering construction realizes the saving by representing many component strings with label rate $I_c = I(H; XY)$, followed by a small exact probability correction. This is the classical form of the ensemble mechanism already identified in [2]; we make no claim to invent that general coding principle.

The elementary phase minimization also extends beyond one table. For arbitrary positive real amplitudes, let

$$B_{\mathrm{gen}}(z, w) = \begin{pmatrix} a & b & 0 \\ 0 & c & d \\ e & fz & gw \end{pmatrix}.$$

Before trace normalization, direct multiplication gives

$$\begin{aligned} e_2(B_{\mathrm{gen}} B_{\mathrm{gen}}^*) &= C_2 - abef(z + z^{-1}) - cdfg(z/w + w/z), \\ \det B_{\mathrm{gen}} &= bde + acg\, w - adf\, z, \end{aligned}$$

where C_2 is independent of the phases. The determinant of the Gram is $|\det B_{\mathrm{gen}}|^2$. Consequently every real affine combination of its second and third symmetric coefficients has the three-cosine form (18). Section 4.4 applies throughout this seven-parameter amplitude family. Useful quadrature rectangles and multipliers still have to be found for a new table.

This structure also gives a search heuristic with a precise limitation. The second coefficient prefers $z = w = 1$. When $bde > acg + adf$, the determinant magnitude instead prefers $z = 1, w = -1$; for our table these three positive terms are $1080, 144, 512$. The competing preferences suggest where nontrivial phase optimization may occur. They alone do not imply nonadditivity; the certified defect (35) does.

The binary obstruction is stronger than an unsuccessful numerical search. Appendix A proves that, for every binary classical–quantum target ρ and every finite ensemble $\rho = \sum_j p_j \tau_j$,

$$E_{\mathrm{P}}(\rho) - S(\rho) \leq \sum_j p_j [E_{\mathrm{P}}(\tau_j) - S(\tau_j)]. \tag{36}$$

Even arbitrary quantum components cannot make the one-copy-cost version of the ensemble bound smaller than $E_{\mathrm{P}}(\rho)$. This rules out the present mechanism for a binary target. It does not identify a convex envelope with $E_{\mathrm{P}}^{\infty} - S$, nor exclude improved regularized component costs or direct collective purifications. The binary additivity question and the choice $d_{\min} \in \{2, 3\}$ therefore remain open.

8 Reproducibility and the reasoning route

The rational certificates and standard-library verifiers accompany this local draft. A permanent repository URL or DOI for the actual certificate package is still pending and is required before public release; a standalone PDF does not supply those files. From the package root, run

```
python -B verify_improvements.py
python -B verify_positive_family.py
python -B verify_package.py --math
```

The first command freshly runs the base verifier, which constructs $B(z,w)B(z,w)^*$ by exact rational Laurent arithmetic, derives the principal-minor coefficients and determinant, and checks $\det(BB^*) = \det(B)\overline{\det(B)}$. It verifies tails, quadrature, outside-box comparisons, eight Laurent identities and positive LDL factorizations, and the ensemble spectral enclosures. It then derives and minimizes all three-cosine remainders and checks the rational finite-copy constants. The second command verifies the positive-family constants against the strengthened lower-bound record and current inputs. The last checks all package hashes and historical graph/event records and runs both mathematical verifiers afresh.

No optimizer or numerical eigensolver is required for these replays. The optional symbolic checks use SymPy, NumPy and SciPy and are supplied separately. The written rank reduction, quadrature implications, direct covering proof, and binary obstruction are mathematical arguments distinct from the finite arithmetic checks. The enormous accepted codebook is specified by a terminating search, not included as an enumerated artifact.

The discovery arose through Recursive Reasoning Exploration (RRE) [5]. Earlier branches studied norm compression and binary purification additivity. The support-degree lemma opened a finite phase description, first used for the uniform 7×7 Fano incidence table. A subsequent exact eight-box refinement proved $E_{\mathrm{P}}(P_F) \geq 1.4852$, while its ensemble upper bound is 1.484125091383097297. That earlier example and its proof records are retained as context in the replay. The present result reduces the local alphabet to three; it does not claim a larger absolute gap than the incidence example.

The subsequent smaller-alphabet search used an exact support catalog containing ten connected 3×3 support classes up to independent row and column permutations and transposition. Under row and column permutations alone there are thirteen classes. The numerical search stopped after 36 weighted tables on two supports produced a candidate. A small edge was removed and amplitudes were rounded to integers, producing (1) with only two free phases. A single coefficient box and a coarser quadrature failed to certify the target. The eight-box partition and finer quadrature succeeded. The remaining support classes and the prepared 4×4 search were not run; this was not an exhaustive search over probabilities.

The three views in `replay/viewer.html` serve different purposes:

View	What the reader can inspect
All math routes	Recorded targets, experiments, rejected probes and completed results across the atlas snapshots.
Proof route	The historical dependencies of the ternary bounds and positive family, including the original Chen–Winter input.
Provenance	The initial graph, inherited results, original source paths and evidence hashes.

The small-alphabet search begins at atlas snapshot v023 and its completed proof is v024. The paper revisions are recorded separately. The inherited atlas ends at v025; the present direct construction, remainder refinement and binary proof are documented in this revision and its source notes. The replay landing page distinguishes these additions from the historical graph. Chronological adjacency is not a logical implication, and numerical experiments are not premises of the lower bound. The replay records mathematical statements, dependencies, and supporting evidence.

9 A visual guide to the reasoning route

Figures 1–3 show the three views of the accompanying mathematical route map. As in the RRE representation [5], stations distinguish questions, discovery evidence, proved claims, and imported results. Their order is logical: the map is a retrospective guide to selected arguments, while the event replay in Section 8 retains the historical record.

The *All math routes* view places the ternary proof alongside its numerical discovery and the unresolved binary branches. The earlier Fano example supplies context. The search suggests a table, but the global lower bound uses the complete phase reduction and exact certificates. The binary compression route remains open; the binary ensemble obstruction excludes only the method stated in Appendix A, not every collective construction.

The *Proof route* view emphasizes two jointly required branches. The support-degree reduction, entropy integral, rectangle certificates, and analytic remainders give the one-copy lower bound. The rational ensemble and direct covering construction give the upper bounds; fixed parameters yield the finite-copy result, while vanishing overhead yields the regularized bound. The positive family additionally uses the full-support reduction, continuity, and an ancillary flagged-mixture construction. The dotted Chen–Winter connections describe the older proof route and are dimmed in this view.

The *Provenance* view distinguishes literature inputs and results recorded in earlier research streams. Inherited tags describe the source record, rather than an additional proof or review status. The historical graph ends at v025; the stations marked r7 identify subsequent mathematical additions retained in this revision. The visualization summarizes the written arguments and their finite certificate inputs; it is not itself a proof certificate.

The figures retain the active tab, legend, and full map. The accompanying package includes an interactive map and an animation, with explanations for individual stations. The package location is discussed in Section 8.

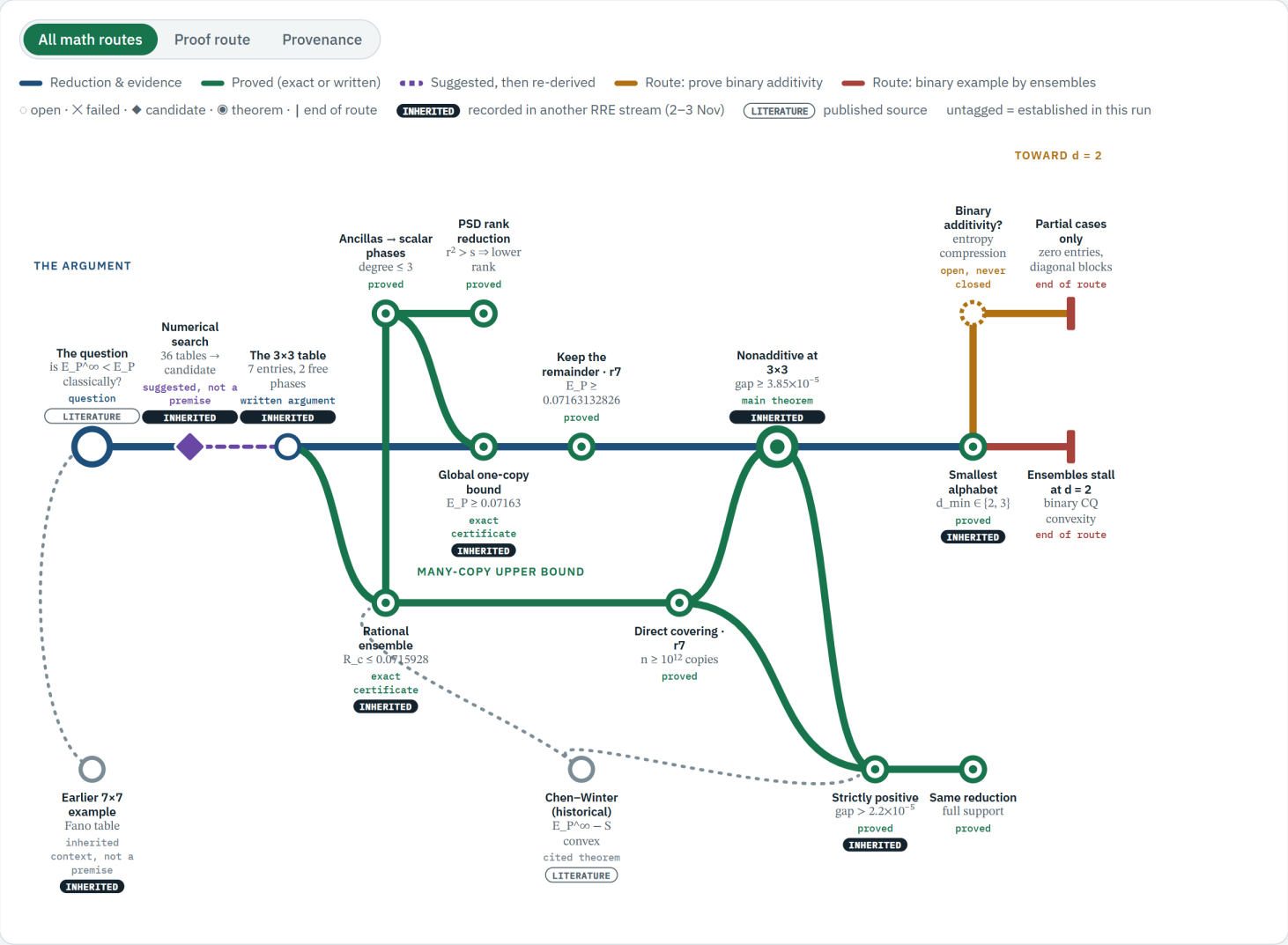


Figure 1: **All math routes.** The ternary argument and selected discovery and binary branches. The numerical candidate motivates the table; the lower and upper proofs jointly establish nonadditivity. The binary question remains open.

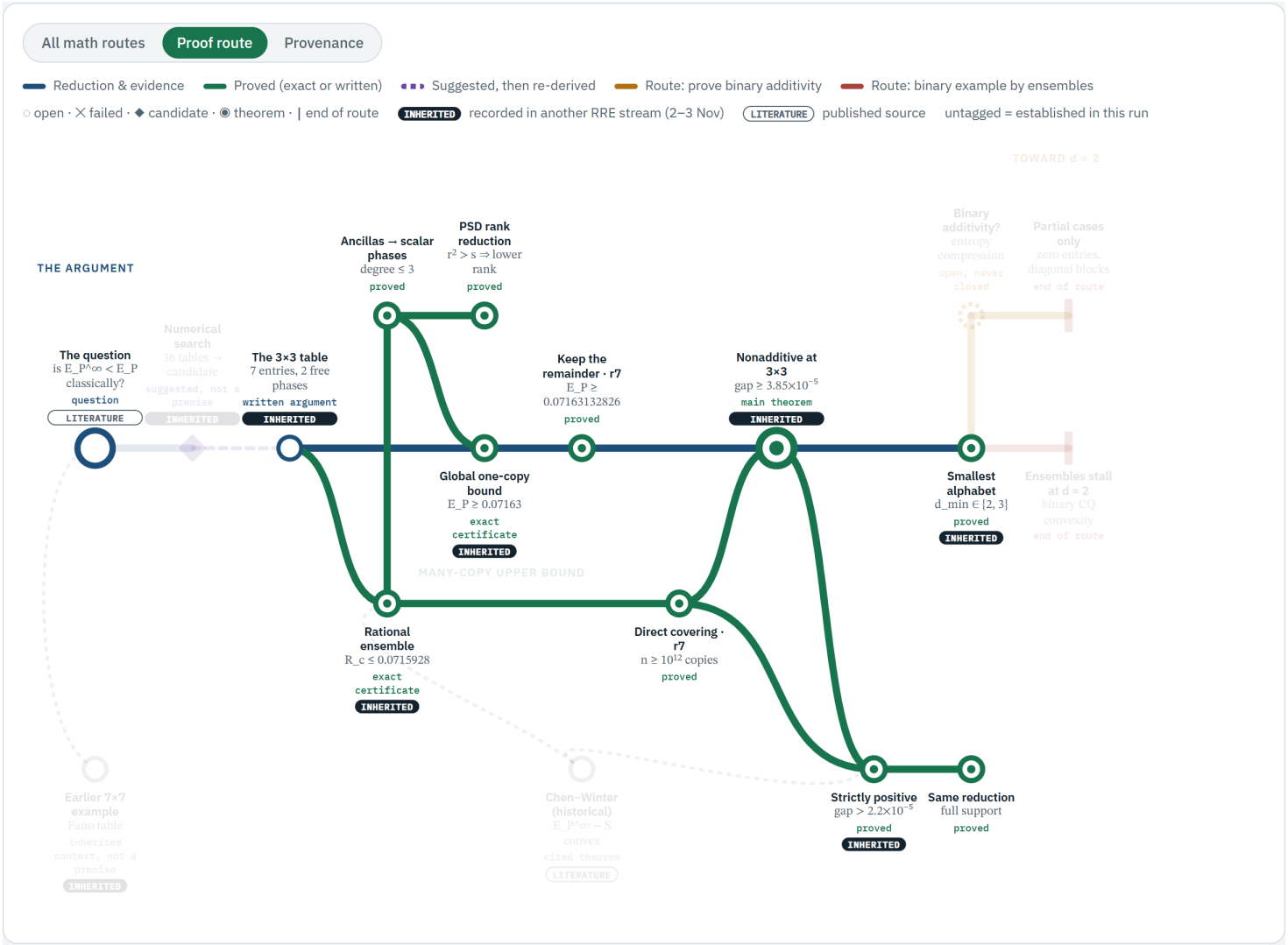


Figure 2: **Proof route.** The complete one-copy lower bound meets the direct many-copy upper construction at the theorem. Chen–Winter is historical context. The positive-family branch adds continuity and the flagged-mixture argument.

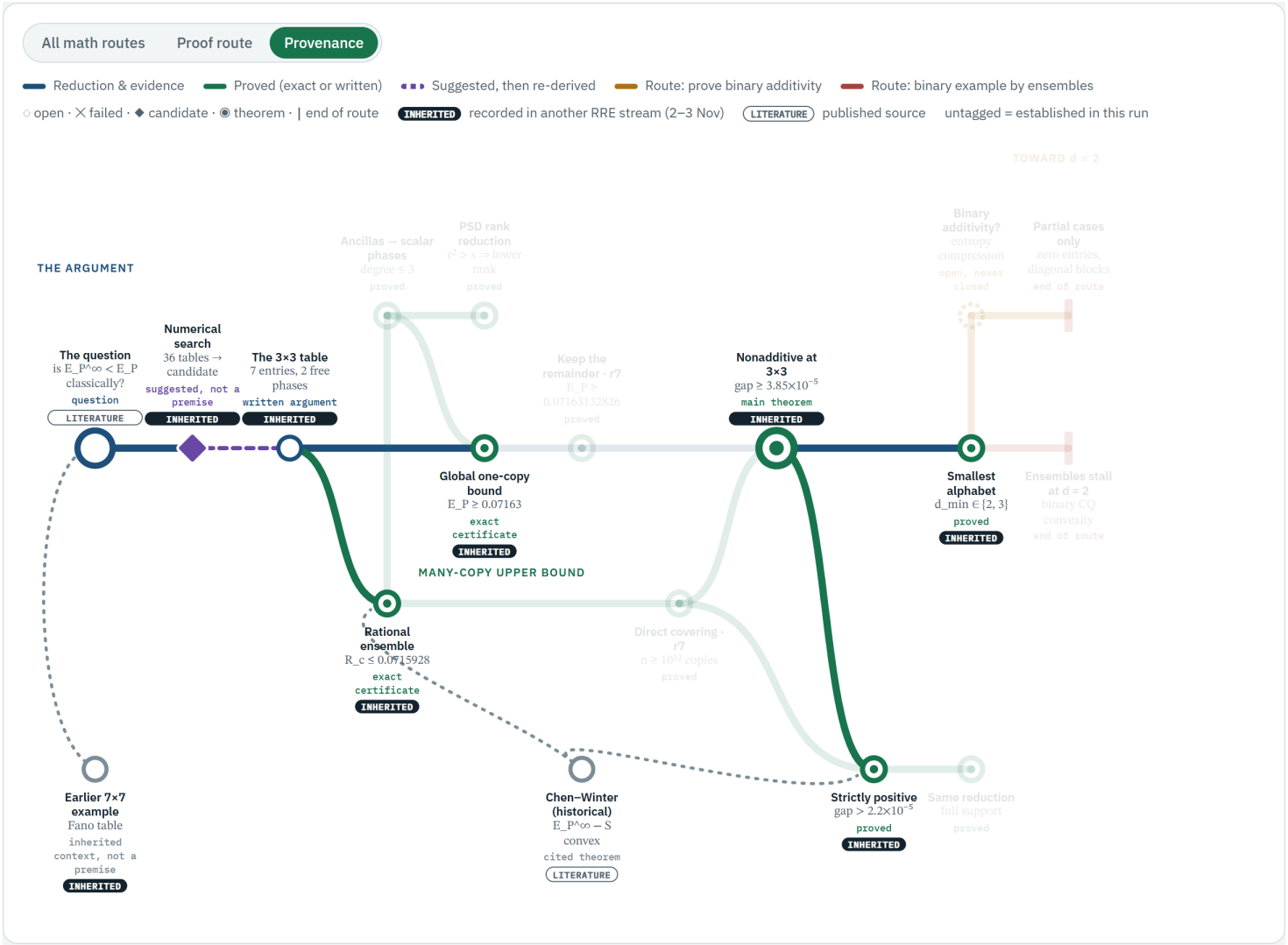


Figure 3: **Provenance.** Literature and inherited research inputs are highlighted. Source labels distinguish the historical graph from the later remainder and covering arguments; they do not assert independent verification or settle the binary case.

A The binary ensemble obstruction

This appendix proves (36) by scalar convexity, fixed-basis quantum convexity, and local dephasing. Its conclusion is separate from Theorem 1. The scalar argument uses unnormalized entropy $\mathcal{S}(X) = -\operatorname{Tr} X \log X$ and $\mathcal{H}(P) = -\sum P_{ij} \log P_{ij}$.

Lemma 7. *For every nonnegative $2 \times N$ table P, the function*

$$F_{\mathrm{sc}}(P) = \mathcal{S}(CC^T) - \mathcal{H}(P), \qquad C_{ij} = \sqrt{P_{ij}},$$

is convex on the full cone, without a normalization constraint.

Proof. The function is homogeneous of degree one. At a strictly positive, rank-two point we can therefore scale the table to total mass one. Write its rows as x_j, y_j, set $c_j = \sqrt{x_j y_j}$, and let

$$G = CC^T = \begin{pmatrix} a & b \\ b & d \end{pmatrix}, \qquad a + d = 1.$$

For an arbitrary affine direction, write $x'_j = x_j u_j$, $y'_j = y_j v_j$, and $\mathbf{z}_j = (u_j, v_j)^T$. In coordinates (X_{00}, X_{01}, X_{11}) for symmetric matrices, define

$$K_j = \begin{pmatrix} x_j & 0 \\ c_j/2 & c_j/2 \\ 0 & y_j \end{pmatrix}, \qquad D_j = \begin{pmatrix} x_j + kc_j & -kc_j \\ -kc_j & y_j + kc_j \end{pmatrix}, \quad k = \tfrac{1}{2}(\log G)_{01} > 0.$$

Let $\mathcal{W}$ be the positive metric defined by $\mathbf{z}^T \mathcal{W} \mathbf{z} = \operatorname{Tr}(X \, D \log_G[X])$. Since $G''_{01} = -\frac{1}{4}\sum_j c_j (u_j - v_j)^2$ and its diagonal second derivatives vanish, differentiation gives

$$F''_{\mathrm{sc}} = \sum_j \mathbf{z}_j^T D_j \mathbf{z}_j - \left(\sum_j K_j \mathbf{z}_j \right)^T \mathcal{W} \left(\sum_j K_j \mathbf{z}_j \right). \tag{37}$$

A Schur complement reduces nonnegativity to $\mathcal{Q} = \mathcal{W}^{-1} - \sum_j K_j D_j^{-1} K_j^T \geq 0$. The vector $(1, 0, 1)^T$ is in its kernel: both terms send it to $(a, b, d)^T$, using $D \log_G[G] = I$ and $D_j (1, 1)^T = (x_j, y_j)^T$. It suffices to restrict to the columns of

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Put $R = \sqrt{(a-d)^2 + 4b^2}$, $\lambda_\pm = (1 \pm R)/2$, and let $\mathcal{L} = (\lambda_+ - \lambda_-)/(\log \lambda_+ - \log \lambda_-)$ be their logarithmic mean. Then $k = b/(2\mathcal{L})$. The inverse logarithmic derivative is $\Omega_G(Y) = \int_0^1 G^s Y G^{1-s} \, ds$. A coordinate covector $\mathbf{z}$ has trace-dual matrix $\left(\begin{smallmatrix} z_0 & z_1/2 \\ z_1/2 & z_2 \end{smallmatrix}\right)$. In an eigenbasis, Ω_G multiplies the diagonal entries by λ_+, λ_- and the off-diagonal entries by $\mathcal{L}$. Consequently, with $D_0 = \operatorname{diag}(1, 1/4)$ and $\mathbf{b} = (2b, -(a-d)/2)^T$,

$$V^T \mathcal{W}^{-1} V = D_0 - \frac{1 - 2\mathcal{L}}{R^2} \mathbf{b}\mathbf{b}^T. \tag{38}$$

For example, the traceless dual matrix for (α, β) has Bloch coordinates $(\beta/2, 0, \alpha)$. Its inverse-metric coefficients are 1 along the Bloch vector of G and $2\mathcal{L}$ perpendicular to it; the perpendicular component is $[2b\alpha - (a-d)\beta/2]/R$, giving (38).

Set $\vartheta_j = \sqrt{x_j/y_j}$, $w_j = y_j$, $d(\vartheta) = k(\vartheta^2+1)+\vartheta$, and $\mathbf{v}(\vartheta) = (2\vartheta, -(\vartheta^2-1)/2)^T$. Direct inversion of the two-by-two D_j gives

$$V^T K_j D_j^{-1} K_j^T V = w_j \left[(\vartheta_j^2+1) D_0 - \frac{k\mathbf{v}(\vartheta_j)\mathbf{v}(\vartheta_j)^T}{d(\vartheta_j)} \right]. \tag{39}$$

Here $\sum_j w_j(\vartheta_j^2+1) = 1$, $\sum_j w_j \mathbf{v}(\vartheta_j) = \mathbf{b}$, and $\sum_j w_j d(\vartheta_j) = k+b$. Weighted Cauchy–Schwarz and (38)–(39) yield

$$V^T \mathcal{Q} V \geq \left[\frac{1}{1+2\mathcal{L}} - \frac{1-2\mathcal{L}}{R^2} \right] \mathbf{b}\mathbf{b}^T \geq 0.$$

The last inequality is equivalent to $\mathcal{L}^2 \geq \lambda_+\lambda_-$. Writing $\lambda_+/\lambda_- = e^{2u}$ proves this from $\sinh(u)/u \geq 1$. Thus the Hessian is nonnegative in the positive rank-two interior.

On an affine segment of strictly positive tables, $\det G$ is analytic on a neighborhood of the segment. If it is not identically zero, its zeros are finite and isolated. At such a zero the small eigenvalue is nonnegative analytic and vanishes to even order at least two. Hence $-\lambda_- \log \lambda_-$ and its first derivative extend continuously, so the derivative of F_{sc} remains nondecreasing across these points. If $\det G$ vanishes identically, put $s = \sum_{ij} P_{ij}$. Then $F_{\mathrm{sc}} = \sum_{ij} P_{ij} \log P_{ij} - s \log s$, with second derivative $\sum_{ij} (P'_{ij})^2/P_{ij} - (s')^2/s \geq 0$. Finally add a common positive table to the two endpoints and let its scale tend to zero. Continuity extends convexity to the entire cone. □

Lemma 8. *For positive semidefinite matrices R_0, R_1, define root fidelity by*

$$f(R_0, R_1) = \operatorname{Tr} \sqrt{\sqrt{R_0} R_1 \sqrt{R_0}},$$

and set

$$G_R = \begin{pmatrix} \operatorname{Tr} R_0 & f(R_0,R_1) \\ f(R_0,R_1) & \operatorname{Tr} R_1 \end{pmatrix}.$$

Then $J(R_0, R_1) = \mathcal{S}(G_R) - \mathcal{S}(R_0) - \mathcal{S}(R_1)$ is jointly convex.

Proof. Fix an orthonormal basis U, let Δ_U be rank-one pinching, and form the two-row table P_U of diagonal entries of R_0, R_1. The function

$$J_U = F_{\mathrm{sc}}(P_U) + D(R_0 \| \Delta_U R_0) + D(R_1 \| \Delta_U R_1)$$

is convex by Lemma 7 and standard joint convexity of quantum relative entropy. Expanding the relative entropies shows that J_U replaces the off-diagonal entry of G_R by $f_U = \sum_i \sqrt{(R_0)_{ii}(R_1)_{ii}}$. The variational expression for root fidelity, the triangle inequality, and Cauchy–Schwarz after inserting the basis projections give $f_U \geq f$. At fixed two-by-two diagonal, entropy decreases as this positive off-diagonal entry increases; hence $J_U \leq J$.

For positive definite R_0, R_1, let

$$X = R_0^{-1/2} (R_0^{1/2} R_1 R_0^{1/2})^{1/2} R_0^{-1/2}.$$

Then $X R_0 X = R_1$ and $\operatorname{Tr}(R_0 X) = f(R_0, R_1)$. Choose U diagonalizing X. If its eigenvalues are $x_i > 0$, then $(R_1)_{ii} = x_i^2 (R_0)_{ii}$, so $f_U = \sum_i x_i (R_0)_{ii} = f$. Thus $J = \sup_U J_U$ is convex. Adding a common ϵI and passing to the limit covers semidefinite matrices. □

Theorem 9. *If $\rho = |0\rangle\langle 0| \otimes R_0 + |1\rangle\langle 1| \otimes R_1$ is normalized, every finite ensemble $\rho = \sum_j p_j \tau_j$ satisfies (36).*

Proof. The exact binary-CQ formula [3, Appendix B, Proposition 1] gives $E_{\mathrm{P}}(\rho) = S(G_R)$. To compare conventions, if $R_0 = p\rho_0$ and $R_1 = (1-p)\rho_1$, then our root fidelity is $f(R_0, R_1) = \sqrt{p(1-p)F(\rho_0, \rho_1)}$, where F is the squared fidelity used in [3]. Zero-weight cases follow by continuity. Its proof constructs a fidelity-saturating purification and bounds the largest marginal eigenvalue of any other purification by $\lambda_{\max}(G_R)$; the two-entry spectrum then majorizes all competitors. Hence $E_{\mathrm{P}}(\rho) - S(\rho) = J(R_0, R_1)$, which is convex for a fixed classical flag basis by Lemma 8.

Let Δ dephase Alice in that basis. For every state τ, $E_{\mathrm{P}}(\Delta\tau) \le E_{\mathrm{P}}(\tau)$: apply a local Stinespring isometry to a purification and assign the environment to Alice's ancilla. The pure-state entanglement is unchanged. Also $S(\Delta\tau) \ge S(\tau)$ by entropy concavity, since dephasing is an average of two unitary conjugations. Apply fixed-basis convexity to $\rho = \sum_j p_j \Delta\tau_j$ and then these two inequalities. □

To emphasize the remaining gap, a sufficient route to binary strong additivity would be the entropy compression inequality

$$S(MM^*) \ge S(cc^*) + \sum_{ij} w_{ij} S(M_{ij}M_{ij}^*/w_{ij}), \qquad w_{ij} = \|M_{ij}\|_F^2, \quad c_{ij} = \sqrt{w_{ij}},$$

for the two-by-two blocks of every purification of $\rho_P \otimes \sigma$. Each normalized block then purifies the same σ, and the blocks obey the global purification constraints. This inequality would imply $E_{\mathrm{P}}(\rho_P \otimes \sigma) \ge E_{\mathrm{P}}(\rho_P) + E_{\mathrm{P}}(\sigma)$. It remains unproved for the required noncommuting blocks. The ensemble obstruction above does not establish it.

Acknowledgment

The author thanks GPT-6.0 Astra for carrying out the Recursive Reasoning Exploration (RRE) run [5] that led to the main mathematical result and for drafting the paper. The author also thanks Claude Opus 5.5 for implementing the visualization of the reasoning routes. The author initiated the RRE run, designed the visualization schema, conducted an initial review of mathematical correctness, provided feedback on subsequent RRE runs that yielded further results, and revised the manuscript. The author takes responsibility for the content and final presentation of this paper.

References

[1] B. M. Terhal, M. Horodecki, D. W. Leung and D. P. DiVincenzo, *The entanglement of purification*, J. Math. Phys. 43(9) (2002), 4286–4298. doi:10.1063/1.1498001. arXiv:quant-ph/0202044v3.

[2] J. Chen and A. Winter, *Non-Additivity of the Entanglement of Purification (Beyond Reasonable Doubt)*, arXiv:1206.1307v1 (2012), Theorem 3 and Corollary 4. doi:10.48550/arXiv.1206.1307.

[3] A.-R. Negari and Z. Baghali Khanian, *Rényi Entanglement of Purification Is Non-additive*, arXiv:2608.28897v1 (2026). doi:10.48550/arXiv.2608.28897.

[4] Z. Baghali Khanian and A.-R. Negari, *Entanglement of purification for Werner states: canonical purification and nonadditivity*, arXiv:2610.03630v1 (2026). doi:10.48550/arXiv.2610.03630.

[5] Y. Xie, *Recursive Reasoning Exploration: Mathematical Research through Evolving Reasoning Graphs—case study: An Exact Certificate-Degree Theorem for Two-Term Tensor Decompositions—Phase III of the Math Instruments Program*, Zenodo (2026). doi:10.5281/zenodo.23849757. Also arXiv:2611.02186v1, doi:10.48550/arXiv.2611.02186.

[6] S. Faraji and Z. Baghali Khanian, *Additivity Results for the Rényi-2 Entanglement of Purification*, arXiv:2605.15439v1 (2026). doi:10.48550/arXiv.2605.15439.

[7] S. Faraji, *Rényi-2 entanglement of purification of the symmetric Werner state*, arXiv:2611.00144v1 (2026). doi:10.48550/arXiv.2611.00144.